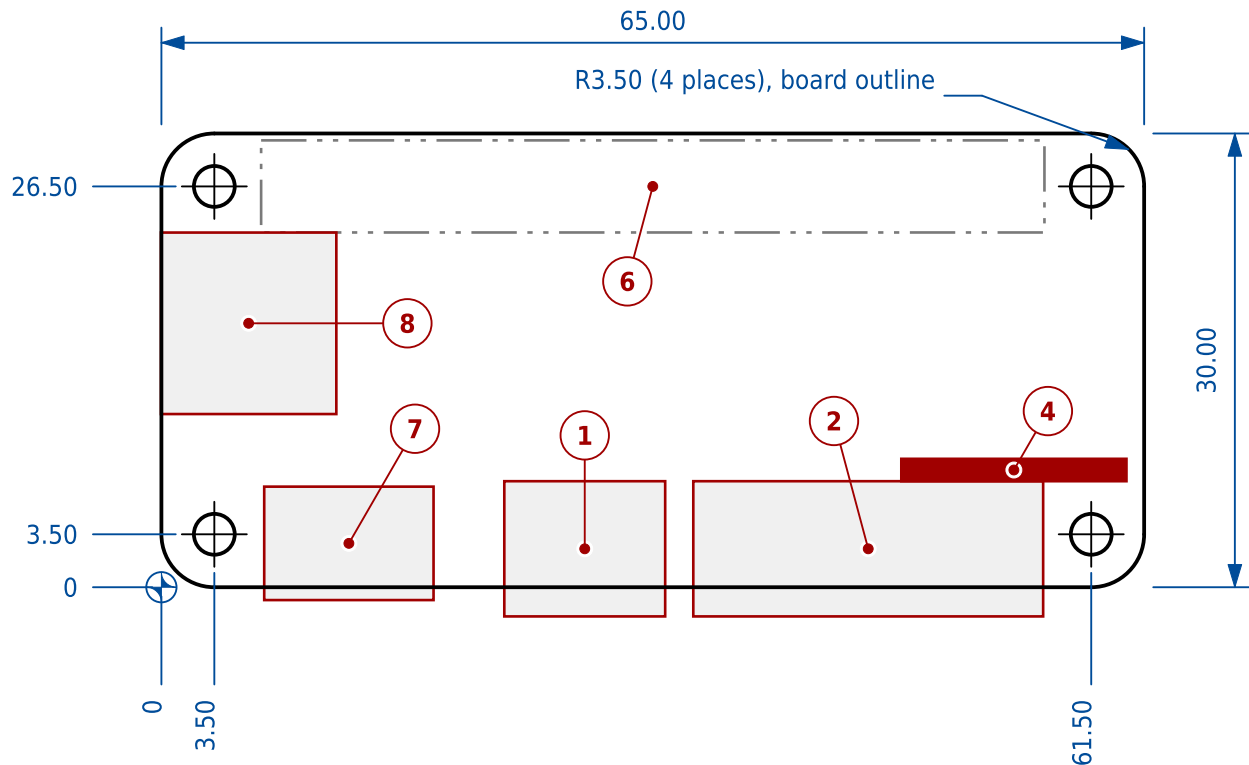




HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
H3	3.50	3.50	2.70	not given
H4	61.50	3.50	2.70	not given
H1	3.50	26.50	2.70	not given
H2	61.50	26.50	2.70	not given

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C J3, JTAG and console	22.68 to 33.32	-1.93 to 7.01
2	USB-C J5 and J4, to the FPGA, 2 on 12.50 mm pitch	35.18 to 58.32	-1.93 to 7.01
3	Ethernet RJ45 - not on this board	-	-
4	User LEDs LED0-LED4, 5 on 2.98 mm pitch	48.94 to 63.83	7.01 to 8.49
5	User LEDs, tri-colour - not on this board	-	-
6	GPIO header J1, 2x20, not fitted	6.59 to 58.41	23.45 to 29.55
7	GPD video connector J2	6.80 to 18.00	-0.85 to 6.65
8	microSD card socket J6	-0.03 to 11.57	11.45 to 23.45
9	Third USB port - not on this board	-	-
10	Fourth USB port - not on this board	-	-
11	Status LEDs, debug controller - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are the KiCad reference designators.
- The mounting holes are the Raspberry Pi Zero pattern: 58.00 x 23.00 mm, 3.50 mm in from each edge of a 65 x 30 board. They are drilled 2.70, the bottom of the Pi Zero drawing's 2.75 +/-0.05 band; the corners are R3.50 where the Pi Zero's are R3.00.
- No Pmod host and no Ethernet jack.
- The unfitted GPIO position is a Raspberry Pi 40-pin header: 1.00 mm holes on a 2.54 mm grid, with the Pi's 5 V, 3V3 and first GPIO pins in the Pi's places and pin 1 at (8.38, 25.23), in the row farther from the board edge.
- The two user buttons are on the underside, centred at (46.40, 15.45) and (46.40, 18.85); a plate under the board has to clear them.
- The board file has been re-annotated since the mass-production release. Nothing drawn here moved, but at d67bb758 the programming port is J5 and the user LEDs run D1 to D5 left to right; the designators here are the ones the sold boards were fabbed with.
- Assembled envelope, connector overhang included: 65.03 x 31.93 mm.

SOURCES

- KiCad board file: https://github.com/cheyao/icepi-zero/hardware/v1.3/icepi-zero.kicad_pcb @ 6e4aaba2 (tag v1.3) - title block: Icepi Zero rev v1.3, dated 2025-07-22; Chengyin Yao (cheyao). Solderpad Hardware Licence 2.1.
- Board revision: <https://github.com/cheyao/icepi-zero/releases/tag/v1.3> - the v1.3 tag, "Final mass production files": the revision made in quantity and sold. The same board file later in its history, at d67bb758, is read too and agrees on every position drawn here.
- Board size: <https://github.com/cheyao/icepi-zero/blob/v1.3/README.md> - quoted at this tag: "Icepi Zero is an FPGA development board in the popular Raspberry Pi Zero form factor"; the board file gives 65.00 x 30.00 mm.
- Assembly: <https://github.com/cheyao/icepi-zero/blob/v1.3/hardware/v1.3/production/bom.csv> - the production bill of materials at the same tag lists no 2x20 header: the GPIO position is not fitted.
- Form factor compared: <https://datasheets.raspberrypi.com/rpizero/raspberry-pi-zero-mechanical-drawing.pdf> - Raspberry Pi Zero, RPI-ZERO-V1_2 of 2015-09-23: 65 x 30 mm, corner radius 3.0, 4x M2.5 mounting holes drilled to 2.75 +/-0.05, 58 x 23 apart and 3.5 in from each edge.

ICEPI ZERO

v1.3, Raspberry Pi Zero form factor

DRAWN		VERSION		UNITS		SCALE	
generated		v0.0-270-gc2d9e2f		mm		2:1	
SIZE	SHEET		DRAWING NO				REV
A3	1 OF 1		FPGA-ICEPI-ZERO				A
MATERIAL			GENERAL TOLERANCE				
PCB, 1.6 nominal			edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				